

Question Paper

Exam Date & Time: 05-Jun-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. DEGREE EXAMINATIONS - JUNE 2025

SUBJECT: ECE 1071/ECE_1071 - BASIC ELECTRONICS

(MAKE-UP)

Marks: 50

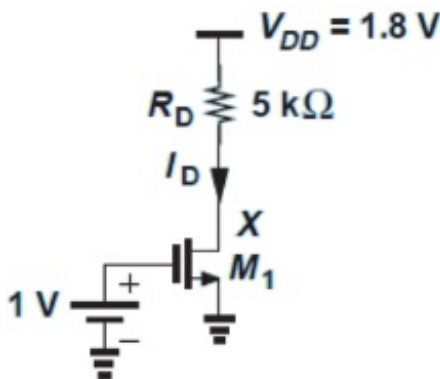
Duration: 180 mins.

Answer all the questions.

- 1A) A Germanium diode has a saturation current of 1pA at 27°C . Determine (a) Diode bias voltage when diode current is 3mA (b) Diode bias current when the temperature changes to 340°K , for the same bias voltage. (c) Also determine the dynamic resistance of the diode at this new temperature. Clearly show all calculation steps (4)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Determine the value of W/L in figure that places M_1 at the edge of saturation. Also, calculate the drain voltage change for a 1-mV change at the gate. Assume $\mu_n C_{ox} = 100\text{ }\mu\text{A/V}^2$ and $V_{TH} = 0.2\text{V}$ (3)



Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Explain the term Peak Inverse Voltage (PIV). How does it affect the selection of diodes in a full-wave and bridge rectifier circuit? (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2A) A sinusoidal voltage with $20\text{-}0\text{-}20\text{ V}$ applied to secondary of the transformer used for full wave rectification. If the load resistance is $1\text{K}\Omega$, determine: (4)
- (a) Peak value of load current
 - (b) Average load current
 - (c) RMS load current
 - (d) Average output voltage
 - (e) Rms output voltage
 - (f) Efficiency
 - (g) Ripple factor
 - (h) PIV

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2B) Design a Zener regulator network with $R_S = 500\Omega$, $V_Z = 12V$ and $V_i = 50V$. Determine the minimum and maximum values of R_L and load current so that Zener diode remains in ON state and safe, given that $I_{ZK} = 2\text{ mA}$ and $P_{ZM} = 500\text{mW}$. Draw the circuit diagram for the same. (3)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2C) Determine CMRR in dB for an OPAMP, if we obtain an output voltage of 5 V for input voltages of $V_1 = 0.5\text{ mV}$ and $V_2 = -0.5\text{ mV}$ applied to non-inverting and inverting terminals respectively, and 10mV if both inputs V_1 and V_2 are 0.5mV. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 3A) Draw the circuit diagram of Op-Amp inverting summing amplifier with V_1 , V_2 as the input voltages and derive the expression for output voltage. Also design the circuit using two Op-Amp's to obtain, $V_{out} = 6V_1 + 0.2V_2 + 4V_3$ where V_1 , V_2 , and V_3 are input voltages. Assume $R_F = 10K\Omega$. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3B) a. Encode the data bits 1111 into 7 bit even parity Hamming code. (3)
b. Identify the corrupted position in the transmitted code 1101111 based on (7,4) even parity Hamming code.
c. Modify the corrupted position and decode the correct data.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3C) Implement a full adder using minimum number of logic gates. Assume A,B and C_{in} are inputs whereas Sum and Carry are outputs. Obtain the Boolean expression for sum and carry using the truth table. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4A) Obtain the minimal sum of product expressions based on the sum of minterms given below using K-map and implement it using NAND gate. (4)
$$F(A,B,C,D) = \sum m(0,2,4,8,10,12) + d(6,7,13,14)$$

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4B) Design a 3 bit Ripple down counter using negative edge triggered T flipflops and draw the output waveforms showing all the possible states of the counter. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4C) Consider a serial input data 10100110 fed to a 4-bit shift register circuit from LSB. Determine the output for SISO operation after 5th clock pulse and the number of clock pulses required to shift MSB bit to the output. Also, draw the circuit diagram. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5A) Given an FM signal $V_{FM}(t) = 20 \cos [2\pi 10^8 t + 5 \sin(2\pi 25000 t)]$, Calculate carrier frequency, modulating frequency, frequency deviation and bandwidth using Carson's rule. (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5B) Draw the block diagram of basic communication System and explain the function of each block. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

5C) For binary data 11010, sketch the ASK, FSK and PSK signals. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name			Subject
MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025			ECE 1071-PHY:BASIC ELECTRONICS
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	7
3	3	CO 3	7
4	4	CO 4	16
5	5	CO 5	10
Total			50

* NA - CO Not Selected

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MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025			ECE 1071-PHY:BASIC ELECTRONICS
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	6(12%)
3	Application	9	31(62%)
4	Analysis	4	13(26%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)